Module B | Examination Review

Four exercise summaries • verified 2026-10-12 estimates • open-ended oral/written practice

One-page map of the four exercises

Task	Research question	Main result
A1 / 1	Which grade category?	Ordinal vs multinomial: PO test p=0.104; ordinal defensible, AIC nearly tied
A1 / 2	Are categorical factors associated?	Log-linear two-way nested LR=417.315 (4 df); equivalence with logistic; goodness of fit p=0.0133
A1 / 3	How many absence hours?	NB2 AIC 4,423.99; more parsimonious than ZINB2
A2	What explains math scores across clusters?	9,623 pupils; 400 schools; 757 classes; school/class ICC 5.45%/15.52%

Assignment 1 / E1 — categorical grades

Multinomial: log[P(Y=k)/P(Y=reference)]=alpha_k+beta_k X; a different beta for each category comparison. **Ordinal proportional odds:** log[P(Y>j)/P(Y≤j)]=alpha_j+beta X for every cutoff j, with a shared beta. Grades: Low=0-9, Medium=10-14, High=15-20. Proportional odds null: all cumulative-logit slopes equal; joint Brant-type Wald chi-square(5)=9.132, p=0.104; no rejection at 5%, not proof. AIC(MN)=764.109 vs AIC(ordinal)=765.309; BIC more strongly favors ordinal. Failures beta=-0.929 gives exp(beta)=0.395 odds multiplier for being in a higher category.

Assignment 1 / E2 — contingency tables and equivalence

Sex × Race × Education (2×5×3; n=32,561). A hierarchical Poisson log-linear specification [SR][SE][RE] matches conditional logistic Sex ~ Race+Education when predictor-predictor interaction Race×Education is unrestricted. The seven corresponding male-level terms match numerically. Nested LR test from conditional independence to full two-way gives chi-square(4)=417.315 (p very small). However, two-way deviance=19.297 on 8 df (p=0.01335): better than simpler models, but not a satisfactory perfect overall fit at 5%.

Assignment 1 / E3 — overdispersed count outcomes

Mean absence 6.924, variance 177.716, 44 zeros out of 740. Poisson imposes Var=mean and fits poorly (AIC 10,133.25). NB2 uses Var=mu+alpha*mu² and is preferred (AIC 4,423.99); ZINB2=4,425.99 and ZIP=9,755.49. NB2 predicts approximately 106 zeros (versus 44 observed), which is a limitation. For zero-inflated models, one regression describes latent count intensity; another describes structural-zero membership odds. Excessive raw count variance does not by itself prove zero inflation.

Assignment 2 — key equations and verified estimates

Three-level model: Y_ijk = X_ijk beta + u_school,k + v_class,jk + e_ijk. Gaussian outcome WLE_MATH_200 is continuous. Fixed effects adjust for observed student and contextual variables; independent random intercepts account for within-school and within-class correlation. All three fitted models converged; birth month/year must be treated as categorical variables according to the data dictionary.

Parameter	Verified estimate	Interpretation
Female (vs male)	-10.197, p<0.001	Lower conditional math score
First-generation origin	-7.493, p=0.0018	Lower than native students
Second-generation origin	-6.622, p<0.001	Lower than native students

Parameter	Verified estimate	Interpretation
ESCS_student	+3.302, p=0.00038	Positive conditional association
ESCS_class	+1.800, p=0.223	Not statistically significant
Geo area 5 (vs 1)	-5.513, p=0.0158	South and Islands lower than North-West
School variance / class variance	66.234 / 122.458	Distinct variance components
Same school / same class ICC	5.45% / 15.52%	Includes nested class variance

Model comparison must use **ML** when the fixed-effect formulas differ: reduced AIC 95,670.15 vs full AIC 94,963.40; LR chi-square(40)=786.746, p=2.50×10^-139. Final variance and fixed estimates can be reported using REML. Cleaning retained 9,623/16,244 records; geographic retention differs (50.7%–65.1%), so generalizability is limited. Fits are unweighted despite original two-stage stratified sampling.

Practice: five questions and short model answers

Q. Why does a multilevel model outperform ordinary regression?

A. It allows observations within the same campus and classroom to share random intercepts, representing the dependence ignored by independent-error regression.

Q. What is the difference between ML and REML?

A. ML uses the full-data likelihood and supports comparison of distinct fixed-effect formulas. REML adjusts for estimation of fixed effects and is suitable for variance components when the fixed specification is held fixed.

Q. How do you interpret ESCS_student = 3.302?

A. One additional unit of the student ESCS index is associated with about 3.30 higher expected math-score units, conditional on the fixed predictors and random effects.

Q. What is the intraclass correlation for pupils in the same class?

A. (66.234+122.458)/(66.234+122.458+1027.225) = about 0.155, meaning approximately 15.5% of modeled variance is shared by classmates via random intercepts.

Q. Can you claim school or family background causes these gaps?

A. No. The study is observational, the likelihood is unweighted and complete-case selection is substantial; we report conditional associations, not causal effects or nationally calibrated estimates.